MERLIN: Detailed Derivations of the Bilateral Controller, the Simulation Model and the Evaluation Metrics

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Supplementary document to the paper

MERLIN – Multi-arm Experimental Robotics Learning and INtegration

Abstract

This document derives every equation used by the MERLIN bilateral controller and by its evaluation: the external-torque estimator, the master impedance law, the slave admittance and its exact discretisation, the force reflection between arms of different kinematics, the baseline teleoperation laws, the offset-free position MPC and its ADMM solver, the energy tank on the feedback channel, the identification of the servo position loop, the simulation model, the metrics, and the parameter-tuning objectives of the Control Simulator. Each result is stated with its assumptions and proved. Statements that can be checked numerically are tied to the unit tests in `ros_ws/src/controller/test/` of the repository. Section 13 lists what is not proven.

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1 Scope, notation and standing assumptions

The teleoperation pair is a ReactorX 200 master and a ViperX 300S slave with the forearm-roll joint locked. Each arm has $n = 5$ active joints $q = [q_{\text{waist}}, q_{\text{shoulder}}, q_{\text{elbow}}, q_{\text{wrist angle}}, q_{\text{wrist rotate}}]^\top$. The controller runs at 50 Hz, $\Delta t = 0.02\,\text{s}$. Subscripts m and s denote master and slave. For a vector x, $\text{diag}(x)$ is the diagonal matrix, $\odot$ the elementwise product, $\text{sat}_a(x) = \text{clip}(x, -a, a)$ and $\text{dz}_\delta(x) = \text{sign}(x)\max(|x| - \delta, 0)$ act elementwise, and $\mathcal{F}_\alpha\{x\}[k] = \alpha x[k] + (1-\alpha)\mathcal{F}_\alpha\{x\}[k-1]$ is a first-order IIR filter.

Assumption 1 (rigid-body model)**.** For each arm, $M(q) = M(q)^\top \succ 0$ is the joint-space inertia matrix, $C(q, \dot{q})$ the Coriolis matrix built from the Christoffel symbols, and $G(q)$ the gravity torque:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau_m + \tau_{ext}. \tag{1}$$

τ_m is the motor torque and τ_{ext} the *external torque*: everything not in the model, i.e. contact and operator forces, joint friction and model error. The Christoffel construction gives $\dot{M} = C + C^\top$, hence $\dot{q}^\top(\dot{M} - 2C)\dot{q} = 0$.

Assumption 2 (actuation and sensing)**.** Motor torque is $\tau_m = n_\mu \odot k_t \odot 10^{-3} I$, with I the present current in mA (as published by the Interbotix `xs_sdk`), k_t the torque constant in N m/A, and n_μ the number of motors per joint (two for the shoulder of the RX200 and for the shoulder and elbow of the VX300S). The current loop is much faster than 50 Hz. Joint positions are measured; $\dot{q}$ is obtained from them by a backward difference with the message time stamps.

Assumption 3 (slave position loop)**.** Each slave joint is driven by a servo position loop, i.e. it accepts a position command u, not a torque. Seen from u, joint i behaves as a decoupled second-order system with a lumped, slowly varying acceleration disturbance c_i (gravity sag, friction):

$$\ddot{q}_i = \omega_i^2(u_i - q_i) - 2\zeta_i\omega_i\dot{q}_i + c_i. \tag{2}$$

ω_i, ζ_i are identified per joint (Section 9). This is a modelling assumption; it is validated in simulation in Section 9.1.

Sign convention. τ_{ext} is the torque the environment applies *on* the joint. The torque the controller renders on the operator points in the same direction as the environment's torque on the slave, so the operator is pushed back.

2 External-torque estimation

2.1 The static residual

Lemma 1 (static residual)**.** *Define* $\hat{\tau}^{static} := G(q) + C(q, \dot{q})\dot{q} - \tau_m$. *Then* $\hat{\tau}^{static} = \tau_{ext} - M(q)\ddot{q}$.

Proof. Solve (1) for τ_{ext}: $\tau_{ext} = M\ddot{q} + C\dot{q} + G - \tau_m$, and subtract $M\ddot{q}$. □

The static residual is exact only when $M\ddot{q}$ is negligible. On a position-controlled arm the servo loop accelerates the links, and $M\ddot{q}$ appears as a false contact torque. In the simulation study the shoulder inertia is $M_{22} \approx 0.23$–$0.30\,\mathrm{kg\,m^2}$; the free-motion false torque at the shoulder is 0.25, 0.48 and 1.08 N m RMS at one, two and three times the nominal motion speed, and its correlation with $-M\ddot{q}$ is 0.67, 0.81 and 0.92. At low speed friction, mass mismatch and current noise dominate; at high speed $M\ddot{q}$ does.

Numerical check: `test_static_residual_equals_tau_ext_minus_Mqdd, test_inverse_dynamics_decomposition`.

2.2 Generalised-momentum observer

Define the generalised momentum $p := M(q)\dot{q}$ and $\beta(q,\dot{q}) := G(q) - C(q,\dot{q})^\top \dot{q}$.

Lemma 2 (momentum balance)**.** *Under Assumption 1, $\dot{p} = \tau_m + \tau_{ext} - \beta(q,\dot{q})$.*

Proof. $\dot{p} = \dot{M}\dot{q} + M\ddot{q}$. Substituting (1), $M\ddot{q} = \tau_m + \tau_{ext} - C\dot{q} - G$, and $\dot{M} = C + C^\top$: $\dot{p} = (C + C^\top)\dot{q} + \tau_m + \tau_{ext} - C\dot{q} - G = \tau_m + \tau_{ext} + C^\top\dot{q} - G$. □

Theorem 3 (observer)**.** *Let $K_O = \mathrm{diag}(k_1,\dots,k_n)$, $k_i > 0$, and define*

$$r(t) = K_O\Big(p(t) - p(0) - \int_0^t \big(\tau_m + C^\top\dot{q} - G + r\big)\,ds\Big). \tag{3}$$

Then $r(0) = 0$ and $\dot{r} = K_O(\tau_{ext} - r)$. In particular r_i is a first-order low-pass filter of $\tau_{ext,i}$ with cut-off k_i rad/s, $r_i(t) = \int_0^t e^{-k_i(t-s)}k_i\,\tau_{ext,i}(s)\,ds$, and $r \to \tau_{ext}$ for a constant τ_{ext}. No acceleration is needed.

Proof. $r(0) = 0$ by construction. Differentiating (3), $\dot{r} = K_O(\dot{p} - \tau_m - C^\top\dot{q} + G - r)$. By Lemma 2, $\dot{p} - \tau_m - C^\top\dot{q} + G = \tau_{ext}$. The solution of the scalar linear ODE $\dot{r}_i = k_i(\tau_{ext,i} - r_i)$, $r_i(0) = 0$, is the stated convolution; for constant τ_{ext} it equals $\tau_{ext,i}(1 - e^{-k_i t})$. □

Numerical check: `test_Mdot_equals_C_plus_Ct, test_observer_converges_to_constant_external_torque`.

Remark 4 (bias from model error). If the implementation uses imperfect $\tilde{M}, \tilde{G}, \tilde{C}, \tilde{k}_t$, repeating the proof gives $\dot{\tilde{r}} = K_O(\tau_{ext} + \delta - \tilde{r})$ with

$$\delta = (\tau_m - \tilde{\tau}_m) + (\tilde{\beta} - \beta) + \Delta\dot{M}\,\dot{q} + \Delta M\,\ddot{q}, \qquad \Delta M = \tilde{M} - M.$$

The estimate is the low-passed external torque *plus* a bias from the torque constants, from gravity and Coriolis errors, and from the inertia error times the acceleration ($\Delta M\ddot{q}$, not $M\ddot{q}$). In the simulation the link masses are 8% larger than the controller's model, so the gravity term alone gives $\delta \approx 0.08\,G \approx 0.2\,\mathrm{N\,m}$ at the shoulder and elbow ($G = [-2.3, -3.0]\,\mathrm{N\,m}$ at the reference pose), the order of the 0.25–0.3 N m RMS error measured in the estimation study. On hardware, k_t and the link masses are calibrated with known weights.

Proposition 5 (discrete observer)**.** *Implement (3) at sample time Δt with the rectangle rule, $S[k] = S[k-1] + \Delta t\,(\tau_m[k] - \beta[k] + r[k-1])$ and $r[k] = K_O(p[k] - p[0] - S[k])$. If $p[k] - p[k-1] = \Delta t(\tau_m[k] - \beta[k] + \tau_{ext}[k]) + \varepsilon[k]$, with $\varepsilon[k]$ the quadrature error of the integral of $\dot{p}$, then*

$$r[k] = (1 - K_O\Delta t)\,r[k-1] + K_O\Delta t\,\tau_{ext}[k] + K_O\varepsilon[k]. \tag{4}$$

This is a first-order IIR filter, stable iff $0 < k_i\Delta t < 2$ and monotone iff $k_i\Delta t < 1$. With $K_O = 15$ rad/s and $\Delta t = 0.02$ s, $a = K_O\Delta t = 0.3$, and the equivalent time constant is $-\Delta t/\ln(1-a) = 56$ ms (continuous value $1/K_O = 67$ ms).

Proof. Subtract the update for $r[k-1]$ from the one for $r[k]$: $r[k]-r[k-1] = K_O\big((p[k]-p[k-1]) - \Delta t(\tau_m[k]-\beta[k]+r[k-1])\big)$. Inserting the balance for $p[k]-p[k-1]$ gives $r[k]-r[k-1] = K_O\Delta t(\tau_{ext}[k]-r[k-1]) + K_O\varepsilon[k]$, which is (4). The pole is $1-k_i\Delta t$, inside the unit circle iff $0<k_i\Delta t<2$, and non-negative iff $k_i\Delta t\le 1$. □

Numerical check: `test_discrete_observer_recursion_is_exact_iir (the recursion holds to` 10^{-10} `when` $\varepsilon = 0$`).`

Remark 6 (unfiltered inputs). The implementation feeds the observer the unfiltered velocity and current. An IIR velocity filter with $\alpha = 0.3$ has a mean delay of $(1-\alpha)/\alpha = 2.3$ samples (47 ms), comparable with the observer time constant; with filtered inputs $p = M\dot q$ lags the torque integral, which acts like a negative inertia in the force loop. In simulation this excited a master–slave limit cycle in the force loop. This is an empirical observation, not a theorem.

3 Master impedance law

The RX200 master runs in current mode, so the commanded current is applied as torque. The master law is

$$\tau_{m,cmd} = \hat G_m(q_m) - D_{md}\dot q_m + \gamma\,\tau_f, \tag{5}$$

where τ_f is the feedback torque of Section 5 (it depends on slave data and on $q_m - q_s$ only), and $\gamma\in[0,1]$ is the gate of the energy tank (Section 8). The command is finally saturated at the torque limits of the master.

Proposition 7 (passivity of the master). *Let $\hat G_m = G_m$ and let the operator apply $\tau_h = J_m^\top F_h$. With $V = \frac12\dot q_m^\top M_m\dot q_m$,*

$$\dot V = \dot q_m^\top\tau_h - \dot q_m^\top D_{md}\dot q_m + \gamma\,\dot q_m^\top\tau_f.$$

Proof. $\dot V = \dot q^\top M\ddot q + \frac12\dot q^\top\dot M\dot q = \dot q^\top(\tau_{m,cmd}+\tau_h - C\dot q - G) + \frac12\dot q^\top\dot M\dot q$. By Assumption 1, $\dot q^\top(\frac12\dot M - C)\dot q = 0$. Insert (5); the gravity terms cancel. □

With $\tau_f = 0$ the master is output-strictly passive from τ_h to $\dot q_m$. The last term, $\gamma\,\dot q_m^\top\tau_f = \gamma P_{out}$, is the power the feedback channel delivers to the operator; it is the quantity the energy tank bounds. Gravity compensation is approximate on hardware, so the statement holds up to $\dot q^\top(\hat G - G)$.

Lemma 8 (design rule: no feedback of the master's own current). *Suppose the master's measured torque is its command delayed by one sample and IIR-filtered, $\tilde\tau[k] = \mathcal{F}_\alpha\{\tau_{cmd}[k-1]\}$, and consider a law that adds an estimate of the master's external torque computed from that measurement, $\tau_{cmd} = \hat G - D_{md}\dot q + \hat\tau_{ext,m} + \tau_f$ with $\hat\tau_{ext,m} = \tilde\tau - C\dot q - \hat G$. Then $\tau_{cmd}[k] = \tilde\tau[k] + b[k]$ with $b = \tau_f - D_{md}\dot q - C\dot q$, which has a pole at $z=1$: a constant $b\neq 0$ makes τ_{cmd} ramp by αb per sample until the torque limit.*

Proof. Substitute $\hat\tau_{ext,m}$: the $\hat G$ terms cancel and $\tau_{cmd}[k] = \tilde\tau[k] + b[k]$. With $\tilde\tau[k] = \alpha\tau_{cmd}[k-1] + (1-\alpha)\tilde\tau[k-1]$, the transfer function from b to τ_{cmd} is $\big(1-(1-\alpha)z^{-1}\big)/(1-z^{-1})$, with its pole at $z=1$. For constant b, $\tilde\tau[k] = \alpha(\tilde\tau[k-1]+b) + (1-\alpha)\tilde\tau[k-1] = \tilde\tau[k-1] + \alpha b$. □

Numerical check: `test_master_law_with_own_estimate_has_a_pole_at_one`.

In current mode the measured current is, up to noise and delay, the commanded current, so it carries no information about the operator that the command does not already contain. Any constant bias (a gravity-model error, a sensor offset, the position-channel term) would be integrated. Law (5) has no path from the master's measured current to its command, so the pole does not arise.

4 Slave admittance and its exact discretisation

For each slave joint the compliance offset q_e obeys the desired impedance

$$M_d\ddot{q}_e + D_d\dot{q}_e + K_d q_e = u, \qquad u = \mathrm{dz}_\delta(\hat{\tau}_{ext,s}), \qquad q_{s,\mathrm{ref}} = Sq_m + o + q_e, \tag{6}$$

with design parameters $M_d, D_d, K_d > 0$ and the joint mapping $S = \mathrm{diag}(s_i)$, o from master to slave. With $x = [q_e, \dot{q}_e]^\top$, $A = \begin{bmatrix} 0 & 1 \\ -K_d/M_d & -D_d/M_d \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 1/M_d \end{bmatrix}$, we have $\dot{x} = Ax + Bu$.

Theorem 9 (exact zero-order-hold discretisation)**.** *If u is constant on $[k\Delta t, (k+1)\Delta t)$, then $x[k+1] = A_d x[k] + B_d u[k]$ exactly, with $A_d = e^{A\Delta t}$, $B_d = \int_0^{\Delta t} e^{As} B\, ds$, and*

$$\exp\Bigl(\Delta t \begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix}\Bigr) = \begin{bmatrix} A_d & B_d \\ 0 & 1 \end{bmatrix}. \tag{7}$$

Proof. Variation of constants gives $x(t_{k+1}) = e^{A\Delta t}x(t_k) + \int_0^{\Delta t} e^{A(\Delta t - s)} B\, u[k]\, ds$, and the substitution $s \mapsto \Delta t - s$ turns the integral into $B_d u[k]$. For (7), let $E(t) = \exp(t\bar{A})$ with $\bar{A} = \begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix}$ and write $E(t) = \begin{bmatrix} E_{11} & F \\ 0 & 1 \end{bmatrix}$. From $\dot{E} = \bar{A}E$: $\dot{E}_{11} = AE_{11}$, $E_{11}(0) = I$, so $E_{11} = e^{At}$; and $\dot{F} = AF + B$, $F(0) = 0$, so $F = \int_0^t e^{A(t-s)} B\, ds = \int_0^t e^{As} B\, ds$ [1]. □

Numerical check: `test_zoh_matches_fine_integration (agrees with a` 10^{-12}`-tolerance ODE solve to` 10^{-10}`)`.

Proposition 10 (stability; explicit Euler)**.** *(i) For $M_d, D_d, K_d > 0$ every eigenvalue λ of A has $\mathrm{Re}\,\lambda < 0$, so the eigenvalues $e^{\lambda\Delta t}$ of A_d satisfy $|e^{\lambda\Delta t}| < 1$ for* every *$\Delta t > 0$. (ii) The static deflection is $q_e = u/K_d$. (iii) Explicit Euler, $x[k+1] = (I + \Delta t A)x[k] + \Delta t B u[k]$, is stable iff $|1 + \lambda\Delta t| < 1$ for both eigenvalues.*

Proof. (i) The eigenvalues are the roots of $M_d s^2 + D_d s + K_d = 0$; a real quadratic with coefficients of one sign has both roots in the open left half-plane (Routh–Hurwitz). (ii) Set $\dot{x} = 0$. (iii) The eigenvalues of $I + \Delta t A$ are $1 + \lambda\Delta t$. □

Numerical check: `test_euler_unstable_with_physical_wrist_inertia, test_admittance_static_deflection_is_tau_over_K`.

Remark 11 (why M_d is a design value, not the link inertia). If the physical inertia of the wrist-rotate link, $7.29 \times 10^{-4}\,\mathrm{kg\,m^2}$, were used as M_d with $D_d = 0.4$ and $K_d = 10$, the roots would be $\lambda = -26.3$ and $-522.4\,\mathrm{s^{-1}}$ ($\omega_n = 117\,\mathrm{rad/s}$, $\zeta = 2.34$). At $\Delta t = 20\,\mathrm{ms}$, $|1 + \lambda\Delta t| = 0.47$ and 9.45, so explicit Euler would diverge, while $|e^{\lambda\Delta t}| = 0.59$ and 3×10^{-5}. The design values $M_d = [0.30, 0.40, 0.30, 0.10, 0.04]$, $D_d = [6, 8, 6, 2, 0.8]$, $K_d = [30, 40, 30, 10, 4]$ give $\omega_n = \sqrt{K_d/M_d} = 10\,\mathrm{rad/s}$ and $\zeta = 1$ on every joint, a bandwidth below that of the slave position loop.

Remark 12 (saturation). The implementation saturates $|q_e| \le q_{e,\max}$ and sets $\dot{q}_e = 0$ when the state pushes into the limit (anti-windup). The saturated filter is bounded by construction; Theorem 9 and Proposition 10 describe the unsaturated range.

5 Force reflection between kinematically different arms

Reflecting joint torques one-to-one renders the wrong Cartesian force when the arms differ. At the reference pose the RX200 reaches 0.388 m and the VX300S 0.530 m (ratio 1.37).

Proposition 13 (damped least-squares wrench)**.** *Let $J_s \in \mathbb{R}^{6\times n}$ be the slave end-effector Jacobian and $\lambda > 0$. The wrench*

$$\hat{F}_s = \arg\min_F \ \|J_s^\top F - \hat{\tau}\|^2 + \lambda^2\|F\|^2 = J_s\big(J_s^\top J_s + \lambda^2 I\big)^{-1}\hat{\tau} \tag{8}$$

satisfies $J_s^\top \hat{F}_s = \hat{\tau} - \lambda^2(J_s^\top J_s + \lambda^2 I)^{-1}\hat{\tau}$, i.e. $J_s^\top \hat{F}_s \to \hat{\tau}$ as $\lambda \to 0$ when J_s has full column rank.

Proof. Setting the gradient to zero, $J_s(J_s^\top F - \hat{\tau}) + \lambda^2 F = 0$, so $(J_s J_s^\top + \lambda^2 I)F = J_s\hat{\tau}$. The push-through identity $(J_s J_s^\top + \lambda^2 I)^{-1}J_s = J_s(J_s^\top J_s + \lambda^2 I)^{-1}$ gives (8). Then $J_s^\top \hat{F}_s = J_s^\top J_s(J_s^\top J_s + \lambda^2 I)^{-1}\hat{\tau} = \hat{\tau} - \lambda^2(J_s^\top J_s + \lambda^2 I)^{-1}\hat{\tau}$. □

Five joints cannot produce every wrench, so the 6-dimensional wrench is not unique; (8) selects the minimum-norm one. The feedback torque on the master is

$$\tau_f = k_{of}\, J_m^\top S_f\, \hat{F}_s - k_{om}\, S\, K_{md}\,(Sq_m + o - q_s), \qquad \hat{F}_s \text{ from (8) with } \hat{\tau} = \mathrm{dz}_\delta(\hat{\tau}_{ext,s}), \tag{9}$$

with $S_f = \mathrm{diag}(s_f I_3,\, s_\mu I_3)$. The first term is the force channel, the second the position channel; the factor S maps the slave-side position error back to master joints (it is the transpose of the joint map $q \mapsto Sq$). The implementation uses $\lambda = 0.02$.

Proposition 14 (consistency)**.** *If the two arms have the same Jacobian at their configurations, $S_f = I$ and $\lambda \to 0$, then $J_m^\top \hat{F}_s = \hat{\tau}$: the Jacobian map reduces to joint-by-joint reflection. For two different arms and the same joint torque τ, joint-by-joint reflection produces the master-side force $F_{\mathrm{id}} = (J_m^\top)^+\tau$, whereas the force that produced τ on the slave is $F_s = (J_s^\top)^+\tau$; at the reference pose $\|F_{\mathrm{id}}\|/\|F_s\| \approx 1.3$.*

Proof. The first claim is Proposition 13 with $J_m = J_s$. The ratio follows from $\tau = J^\top F$ (virtual work, $\delta W = \tau^\top \delta q = F^\top J \delta q$): a shorter arm needs a larger force for the same joint torque. □

Numerical check: `test_dls_wrench_reproduces_joint_torque`, `test_identical_arms_give_identity_map`, `test_different_arms_over_render_joint_torque_reflection`.

Remark 15 (power). $\Gamma = J_m^\top S_f J_s^{+\top}$ is not power-preserving between the arms, because the arms are coupled in joint space ($q_m \leftrightarrow q_s$), not in task space. The slave-side power $\hat{\tau}^\top \dot{q}_s = \hat{F}^\top \dot{x}_s$ and the master-side power $\tau_f^\top \dot{q}_m = \hat{F}^\top S_f^\top \dot{x}_m$ are equal only if $\dot{x}_m = S_f^{-\top}\dot{x}_s$, which joint-space mapping does not enforce. This is why the force channel is wrapped by the energy tank of Section 8.

6 Teleoperation architectures compared

All architectures share the estimator of Section 2, the master law (5) and the slave interface. They differ in τ_f and in the slave command u. With $\bar{q}_m = Sq_m + o$ the mapped master position and $\Delta u_{\max}$ the command-rate limit, the rate-limited reference is $u = \mathrm{clip}(\bar{q}_m,\ u_{-1} - \Delta u_{\max},\ u_{-1} + \Delta u_{\max})$.

Architecture	feedback torque τ_f	slave command u
Unilateral	0	rate-limited $\bar{q}_m$
Position–position	$-S\big(K_p(\bar{q}_m - q_s) + K_d(S\dot{q}_m - \dot{q}_s)\big)$	rate-limited $\bar{q}_m$
Position–force	$k_{of}\ \mathrm{dz}_\delta(\hat{\tau}_{ext,s})$ (joint by joint, no tank)	rate-limited $\bar{q}_m$
Proposed without MPC	(9), gated by the tank	rate-limited $q_{s,\mathrm{ref}}$ of (6)
Proposed	(9), gated by the tank	MPC (11)

In the simulation study $K_p = [3, 5, 3, 1, 0.3]$, $K_d = [0.05, 0.08, 0.05, 0.02, 0.005]$, $K_{md} = [3, 5, 3, 1, 0.3]$, $D_{md} = [0.08, 0.12, 0.08, 0.03, 0.01]$, $k_{of} = 0.3$, $k_{om} = 1$, $\delta = [0.15, 0.3, 0.2, 0.1, 0.05]\,\mathrm{N\,m}$, $\Delta u_{\max} = 0.15\,\mathrm{rad}$ per sample, and $q_{e,\max} = 0.3\,\mathrm{rad}$.

7 Slave MPC on the position loop

7.1 Model and prediction

By Theorem 9 with $A = \begin{bmatrix} 0 & 1 \\ -\omega^2 & -2\zeta\omega \end{bmatrix}$ and $B = \begin{bmatrix} 0 \\ \omega^2 \end{bmatrix}$, joint i of (2) gives $x[k+1] = A_d x[k] + B_d u[k]$ for $x = [q, \dot{q}]^\top$ (the disturbance c is handled in Section 7.4). By induction, $x[k] = A_d^k x[0] + \sum_{j=0}^{k-1} A_d^{k-1-j} B_d u[j]$. For one joint, with horizon N, model state $\hat{x}_0$ and output-disturbance estimate $\hat{d}$, the predicted positions and velocities are therefore

$$\hat{q}_k = e_1^\top A_d^k \hat{x}_0 + \sum_{j=0}^{k-1} e_1^\top A_d^{k-1-j} B_d\, u_j + \hat{d}, \qquad \hat{v}_k = e_2^\top A_d^k \hat{x}_0 + \sum_{j=0}^{k-1} e_2^\top A_d^{k-1-j} B_d\, u_j, \qquad k = 1, \dots, N, \tag{10}$$

that is $\hat{q} = \Phi_q \hat{x}_0 + G_q U + \hat{d}\mathbf{1}$ and $\hat{v} = \Phi_v \hat{x}_0 + G_v U$ with $U = [u_0, \dots, u_{N-1}]^\top$, $(\Phi_q)_k = e_1^\top A_d^k$, and G_q, G_v lower-triangular Toeplitz with $(G_q)_{kj} = e_1^\top A_d^{k-1-j} B_d$ for $j < k$.

7.2 The quadratic program

Let r_k be the reference preview, $r_k = q_{s,\mathrm{ref}} + k\Delta t\, S\dot{q}_m$ (constant master velocity over the horizon), $w = (Q, \dots, Q, P)$, $W = \mathrm{diag}(w)$, D the unit lower-bidiagonal difference matrix ($(DU)_k = u_k - u_{k-1}$) and $d_0 = [u_{-1}, 0, \dots, 0]^\top$ with u_{-1} the last applied command:

$$\min_U \ (\hat{q}-r)^\top W (\hat{q}-r) + R\,\|DU - d_0\|^2 \quad \text{s.t.} \quad q_{\min} \le u_k \le q_{\max}, \quad |u_k - u_{k-1}| \le \Delta u_{\max}, \quad |\hat{v}_k| \le \dot{q}_{\max}. \tag{11}$$

Proposition 16 (condensed form)**.** *Let $a = \Phi_q \hat{x}_0 + \hat{d}\mathbf{1} - r$. Then (11) equals $\frac{1}{2} U^\top H U + f^\top U + \mathrm{const}$ subject to $l \le CU \le h$, with*

$$H = 2\big(G_q^\top W G_q + R\, D^\top D\big), \qquad f = 2 G_q^\top W a - 2R\, D^\top d_0, \qquad C = \begin{bmatrix} I \\ D \\ G_v \end{bmatrix}, \tag{12}$$

$$l = \begin{bmatrix} q_{\min}\mathbf{1} \\ d_0 - \Delta u_{\max}\mathbf{1} \\ -\dot{q}_{\max}\mathbf{1} - \Phi_v \hat{x}_0 \end{bmatrix}, \qquad h = \begin{bmatrix} q_{\max}\mathbf{1} \\ d_0 + \Delta u_{\max}\mathbf{1} \\ \dot{q}_{\max}\mathbf{1} - \Phi_v \hat{x}_0 \end{bmatrix}.$$

$H \succ 0$ for $R > 0$, so a feasible problem has a unique minimiser.

Proof. Expand $(G_q U + a)^\top W (G_q U + a) + R(DU - d_0)^\top (DU - d_0)$ and collect the terms quadratic and linear in U. The second constraint group is $(DU)_k \in [(d_0)_k - \Delta u_{\max}, (d_0)_k + \Delta u_{\max}]$, because $(DU)_0 - u_{-1} = u_0 - u_{-1}$ and $(DU)_k = u_k - u_{k-1}$ for $k \ge 1$; the third is $G_v U \in [-\dot{q}_{\max} - \Phi_v \hat{x}_0,\ \dot{q}_{\max} - \Phi_v \hat{x}_0]$. D is unit lower bidiagonal, hence invertible, so $D^\top D \succ 0$; and $G_q^\top W G_q \succeq 0$ since $W \succeq 0$. □

7.3 Solving the QP: ADMM

With the auxiliary variable $Z = CU$, ADMM with proximal weight $\sigma > 0$ and penalty $\rho > 0$ iterates

$$U^+ = (H + \sigma I + \rho C^\top C)^{-1} \big(\sigma U - f + C^\top(\rho Z - y)\big), \tag{13}$$

$$Z^+ = \Pi_{[l,h]}\big(CU^+ + y/\rho\big), \qquad y^+ = y + \rho\,(CU^+ - Z^+), \tag{14}$$

which is the OSQP scheme without over-relaxation [3], a special case of ADMM [4].

Proposition 17 (ADMM)**.** *(i) (13) minimises $\frac{1}{2} U^\top H U + f^\top U + \frac{\sigma}{2}\|U - U^k\|^2 + \frac{\rho}{2}\|CU - Z^k + y^k/\rho\|^2$. (ii) A fixed point (U, Z, y) of (13)–(14) satisfies the KKT conditions of (11): $HU + f + C^\top y = 0$, $Z = CU \in [l, h]$, $y \in N_{[l,h]}(Z)$. (iii) The matrix $H + \sigma I + \rho C^\top C$ does not depend on $\hat{x}_0$, r, u_{-1}, $\hat{d}$ or the bounds, so it is inverted once; an iteration costs one $N \times N$ product and one clipping per joint. (iv) For any $\rho, \sigma > 0$ and a feasible convex QP the iterates converge to a KKT point [4, 3].*

Proof. (i) The gradient condition is $HU + f + \sigma(U - U^k) + \rho C^\top(CU - Z^k + y^k/\rho) = 0$, i.e. (13). (ii) At a fixed point the y-update gives $CU = Z$; the Z-update gives $Z = \Pi(Z + y/\rho)$, i.e. y/ρ lies in the normal cone of $[l, h]$ at Z; and (13) with $U^+ = U^k = U$, $Z^k = Z$ reduces to $HU + \rho C^\top CU = -f + \rho C^\top Z - C^\top y$, which with $CU = Z$ is $HU + f + C^\top y = 0$. (iii) is read off. (iv) is the cited result. □

Iteration budget and choice of ρ. The controller runs a fixed budget of 60 warm-started iterations per tick, so the accuracy of the first move depends on ρ. With $\sigma = 10^{-6}$ and $\rho = 0.3$, the applied first move is within 0.03 mrad on average (0.2 mrad at most) of the exact QP solution over a closed-loop run, and within 2 mrad from a cold start. A large penalty slows convergence in this problem: with $\rho = 10$ the warm-started first move is on average 13 mrad from the exact solution. A solve takes about 0.9 ms for five joints.

Numerical check: `test_admm_solves_the_stated_qp (reference: SLSQP on the same QP data), test_applied_command_always_respects_limits`.

Proposition 18 (the applied command is always admissible)**.** *After the solve, $u_0^\star$ is clipped to $[q_{\min}, q_{\max}]$ and then to $[u_{-1} - \Delta u_{\max}, u_{-1} + \Delta u_{\max}]$. If $u_{-1} \in [q_{\min}, q_{\max}]$ the intersection of the two intervals is non-empty and contains the result, so the applied command satisfies the position and rate limits regardless of solver accuracy. The predicted-velocity bound is enforced only through the QP.*

Proof. u_{-1} belongs to both intervals $I_1 = [q_{\min}, q_{\max}]$ and $I_2 = [u_{-1} - \Delta u_{\max}, u_{-1} + \Delta u_{\max}]$. Let $p \in I_1$ be the result of the first clip and c that of the second, so $c \in I_2$. If $p \in I_2$ then $c = p \in I_1$. Otherwise c is the end point of I_2 nearest to p, which lies between p and u_{-1}; both are in the interval I_1, hence so is c. □

7.4 Offset-free tracking

The servo position loop has no gravity feed-forward, so under a constant load joint i settles at $u + c/\omega^2$ rather than at u. The model (10) without $\hat{d}$ cannot predict this. The controller therefore runs the model in parallel with the plant (*internal-model* form), $\hat{x}[k+1] = A_d\hat{x}[k] + B_d u[k]$, and estimates an output disturbance

$$\hat{d}[k+1] = (1 - \alpha_d)\hat{d}[k] + \alpha_d\big(q[k] - \hat{x}_1[k]\big), \qquad 0 < \alpha_d \le 1, \tag{15}$$

with $\alpha_d = 0.2$.

Proposition 19 (offset-free)**.** *Let the plant be (2) with constant c and $u \equiv u_\infty$. Then $\hat{d} \to c/\omega^2$. Moreover, if the reference is constant, $r \equiv r_\infty$, and $u_\infty = r_\infty - \hat{d}$ lies within the limits, then $(u_\infty,\ \hat{d} = c/\omega^2)$ is an equilibrium of the closed loop at which the plant position equals r_∞.*

Proof. The plant converges to $q_\infty = u_\infty + c/\omega^2$ (A_d is Schur by Proposition 10(i), applied to the loop with $M_d = 1$, $D_d = 2\zeta\omega$, $K_d = \omega^2$); the parallel model converges to $\hat{x}_{1,\infty} = u_\infty$. Hence $q - \hat{x}_1 \to c/\omega^2$, and the stable recursion (15) (pole $1 - \alpha_d$) converges to the same value. For the second claim, with $\hat{d} = c/\omega^2$ and $U \equiv u_\infty = r_\infty - \hat{d}$, the predictions satisfy $\hat{q}_k = r_\infty$ for all k and $DU - d_0 = 0$, so the cost of (11) is 0, its global minimum; by Proposition 16 the minimiser is unique, so the MPC returns u_∞ and the plant settles at $u_\infty + c/\omega^2 = r_\infty$. □

Numerical check: `test_offset_free_estimate_converges_and_removes_steady_state_error (sag 48 mrad removed; $\hat{d}$ within 2%)`.

This is the standard offset-free MPC mechanism [5, 6]. Proposition 19 establishes the zero-error equilibrium, not its asymptotic stability, which is checked in simulation only.

7.5 What is not claimed

No terminal cost or terminal set with a Lyapunov argument is used, so closed-loop stability of the MPC is not proven; $N = 10$, $Q = 100$, $P = 300$ and $R = 0.5$ were chosen in simulation. The prediction model is decoupled per joint.

8 Energy tank on the force channel

Let $P_{out}[k] = \tau_f[k]^\top \dot{q}_m[k]$ be the power the (ungated) feedback channel delivers to the operator–master system, and $P_{in}[k] = \eta\big(\dot{q}_m^\top D_{md}\dot{q}_m + \dot{q}_e^\top D_d \dot{q}_e\big)[k] \geq 0$ $(0 < \eta \leq 1)$ the power the controller dissipates through its two damping terms (master damping and admittance damping). The tank state is $T[k] \in [T_{\min}, T_{\max}]$, $0 < T_{\min} \leq T[0] \leq T_{\max}$, and the gate $\gamma[k] \in [0, 1]$ scales the feedback torque:

$$\gamma[k] = \begin{cases} 1 & \text{if } P_{out}[k] \leq 0 \ \text{ or } \ b[k] \geq \Delta t\, P_{out}[k], \\ \dfrac{b[k]}{\Delta t\, P_{out}[k]} & \text{otherwise,} \end{cases} \qquad b[k] = T[k] + \Delta t P_{in}[k] - T_{\min}, \tag{16}$$

$$T[k+1] = \min\big\{T_{\max},\ T[k] + \Delta t\,(P_{in}[k] - \gamma[k] P_{out}[k])\big\}. \tag{17}$$

The simulation uses $T[0] = 0.5\,\text{J}$, $T_{\min} = 0.05\,\text{J}$, $T_{\max} = 1\,\text{J}$ and $\eta = 1$.

Theorem 20 (bounded delivered energy)**.** *For any signals $\tau_f, \dot{q}_m$ and any $P_{in} \geq 0$, and every $K \geq 1$: (i) $\gamma[k] \in [0, 1]$ and $T[k] \geq T_{\min}$ for all k; (ii) the energy delivered by the gated feedback channel is bounded,*

$$\sum_{k=0}^{K-1} \Delta t\, \gamma[k] P_{out}[k] \ \leq\ T[0] - T_{\min} + \sum_{k=0}^{K-1} \Delta t\, P_{in}[k]. \tag{18}$$

Proof. (i) By induction on $T[k] \geq T_{\min}$. Since $P_{in} \geq 0$, $b[k] \geq 0$, so $\gamma \geq 0$, and $\gamma \leq 1$ by construction. If $P_{out} \leq 0$, then $\gamma = 1$ and $T[k+1] = \min\{T_{\max}, T[k] + \Delta t(P_{in} + |P_{out}|)\} \geq \min\{T_{\max}, T[k]\} \geq T_{\min}$. If $P_{out} > 0$ and $b \geq \Delta t P_{out}$, then $\gamma = 1$ and $T[k] + \Delta t(P_{in} - P_{out}) = b + T_{\min} - \Delta t P_{out} \geq T_{\min}$. Otherwise $\gamma \Delta t P_{out} = b$ and $T[k] + \Delta t P_{in} - b = T_{\min}$. In every case the argument of the min in (17) is $\geq T_{\min}$, and $T_{\max} \geq T_{\min}$. (ii) Write (17) as $T[k+1] = T[k] + \Delta t(P_{in} - \gamma P_{out}) - c[k]$ with the clipping loss $c[k] \geq 0$. Summing over k, $\sum \Delta t \gamma P_{out} = T[0] - T[K] + \sum \Delta t P_{in} - \sum c[k] \leq T[0] - T_{\min} + \sum \Delta t P_{in}$, using $T[K] \geq T_{\min}$ and $c \geq 0$. □

Numerical check: `test_tank_energy_bound_theorem (random signals, 5000 steps)`, `test_energy_tank_never_below_minimum`.

Reading of the result. Whatever the force estimate does, the feedback channel returns to the operator at most the energy the controller has dissipated plus the initial reserve $T[0] - T_{\min}$, and $T_{\max}$ caps how much can be released in a burst. Combined with Proposition 7, the gated master satisfies $\dot{V} \leq \dot{q}^\top \tau_h - \dot{q}^\top D_{md} \dot{q} + \gamma P_{out}$, with the last term energetically bounded by (18). The construction follows the energy-tank architectures of [7, 8].

What it does not say.

- (18) uses the Riemann sum $\Delta t\, \tau_f^\top \dot{q}_m$ for the energy of a sample; it is exact only up to the discretisation error of that product.
- It bounds the energy delivered by the *master-side* controller. With communication delay the slave-side quantities in τ_f are stale, and a delayed channel is not passive in general [10]. A two-port tank or time-domain passivity control with energy exchange between the sites [8, 9] is

required for delayed operation. In the simulation study the tank lowers the worst-case tracking error without delay (from 0.87 to 0.50 rad at $k_{of} = 1.0$, from 1.99 to 0.72 rad at $k_{of} = 1.5$) and gives no consistent benefit at 20–60 ms of delay.

- Passivity of the controller port together with a passive operator and environment yields stability of the interconnection; neither passivity is verified here.

9 Identification of the servo position loop

Given a log of commands $u[k]$ and measured positions $q[k]$ of one joint at times t_k, the parameters $\theta = (\omega, \zeta, c)$ of (2) are estimated by output-error least squares:

$$\hat{\theta} = \arg\min_{\theta} \sum_{k} \big(q[k] - \hat{q}_{\theta}[k]\big)^2, \tag{19}$$

where $\hat{q}_{\theta}$ is the response of (2) to the logged u held constant over each sample (zero-order hold, integrated with a 1 ms semi-implicit Euler sub-step) from the logged initial position at rest. The quality of the fit is reported as $R^2 = 1 - \sum_k (q[k] - \hat{q}_{\hat{\theta}}[k])^2 / \sum_k (q[k] - \bar{q})^2$. On the simulated slave, step tests of 0.15 rad per joint gave $\omega = [25.7, 23.4, 31.1, 98.7, 94.6]$ rad/s and $\zeta = [0.27, 0.30, 0.42, 0.66, 0.46]$. Output-error identification (rather than equation-error regression on $\ddot{q}$) avoids differentiating the encoder signal twice.

9.1 Validation of the prediction model

For a logged run, at every tick k the controller's own model (parallel state and $\hat{d}$, exactly as in the MPC) predicts the slave position h ticks ahead for the commands that were actually applied. The RMS prediction error over three operator seeds was

Predictor	$h = 1$ (20 ms)	$h = 2$ (40 ms)	$h = 5$ (100 ms)	$h = 10$ (200 ms)
hold last position	12.0	23.8	57.6	108.9
constant velocity	2.8	7.8	28.6	71.9
position loop, $\hat{d} = 0$	26.4	26.4	26.5	26.6
position loop with $\hat{d}$ (MPC model)	5.1	6.1	7.5	6.8

RMS error in mrad, pooled over the five joints (simulation).

Without $\hat{d}$ the model is biased by the static sag of the servos; with $\hat{d}$ it is the most accurate predictor from $h = 5$ on, the horizon the MPC uses, whereas constant-velocity extrapolation is better for the first 40 ms.

10 Simulation model

The study uses MuJoCo models of both arms, built from the same robot descriptions (URDF) as the controller, whose dynamics are computed with Pinocchio. Physics runs at $\Delta t_{sim} = 1$ ms and control at $\Delta t = 20$ ms.

- **Model mismatch.** Link masses and inertias are scaled by 1.08 in the plant only. Joint viscous damping $[0.03, 0.05, 0.04, 0.02, 0.01]$ N m s/rad and Coulomb friction $[0.04, 0.08, 0.06, 0.03, 0.01]$ N m are present in the plant and not in the controller.
- **Slave servo.** $\tau_s = \mathrm{sat}_{\tau_{stall}}\left(K_p^s(u - q_s) - K_d^s \dot{q}_s\right)$ with $K_p^s = [60, 150, 100, 25, 6]$ N m/rad, $K_d^s = [1.2, 3.0, 2.0, 0.25, 0.05]$ N m s/rad, stall torques $[10.6, 21.2, 21.2, 10.6, 4.1]$ N m, no gravity feedforward.
- **Master.** The commanded torque is applied directly (current mode), saturated at the stall torques $[4.1, 8.2, 4.1, 4.1, 1.5]$ N m.

- **Sensors.** Positions are quantised to $2\pi/4096$. The reported current of each arm is $I = 10^3\tau/(n_\mu k_t) + \nu$, $\nu \sim \mathcal{N}(0, 5^2)$ mA, quantised to 2.69 mA; for the master, τ is the command of the previous tick, for the slave the servo torque of the last physics step.
- **Operator.** $\tau_h = K_h(q^\star(t) - q_m) - B_h\dot{q}_m$ with $K_h = [6, 10, 6, 2, 0.5]$ N m/rad, $B_h = [0.2, 0.3, 0.2, 0.05, 0.02]$ N m s/rad and an intended trajectory $q^\star$. In the benchmark, $q^\star$ is a sum of sinusoids (amplitudes $[0.6, 0.25, 0.3, 0.4, 0.6]$ rad, 0.25–0.5 Hz) for 12 s, then two raised-cosine pushes of the shoulder towards a wall. For the seed $s > 0$, amplitudes are scaled by $\mathcal{U}(0.75, 1.25)$, frequencies by $\mathcal{U}(0.85, 1.15)$, phases are drawn from $\mathcal{U}(0, 2\pi)$, and the push amplitude and frequency are scaled.
- **Environment.** A wall $x = x_w$ in the slave base frame, 3 cm in front of the start pose, with penalty contact $F_x = -\max\{k\,p + b\,\dot{p}, 0\}$ for penetration $p > 0$, $k = 2000$ N/m (varied from 500 to 10 000), $b = 30$ N s/m.
- **Delay.** For a one-way delay of d ticks, the master-side and the slave-side controllers run as separate instances, each receiving the other arm's measurements d ticks late.
- **Force felt by the operator.** $f_{felt} = \arg\min_f \|J_m^\top f - (\tau_{m,applied} - G_m^{true})\|$: the Cartesian force equivalent to the net non-gravitational torque of the master motors.

11 Metrics

11.1 Tracking

For errors $e[k, j] = q_s[k, j] - q_m[k, j]$ over K samples and J joints, $\mathrm{RMSE}_j = \sqrt{\frac{1}{K}\sum_k e[k, j]^2}$, $\mathrm{MAE}_j = \frac{1}{K}\sum_k |e[k, j]|$, $\mathrm{Max}_j = \max_k |e[k, j]|$, and the *pooled* RMSE is $\mathrm{RMSE}_{\mathrm{pool}} = \sqrt{\frac{1}{KJ}\sum_{k,j} e[k, j]^2} = \sqrt{\frac{1}{J}\sum_j \mathrm{RMSE}_j^2}$.

Proposition 21. $\frac{1}{J}\sum_j \mathrm{RMSE}_j \le \mathrm{RMSE}_{\mathrm{pool}}$, *with equality iff all* RMSE_j *are equal.*

Proof. By Jensen's inequality (convexity of $x \mapsto x^2$), $\left(\frac{1}{J}\sum_j a_j\right)^2 \le \frac{1}{J}\sum_j a_j^2$ for $a_j = \mathrm{RMSE}_j$, with equality iff all a_j coincide. □

For the hardware trials of the paper (Table I), the per-joint values 0.2250, 0.2715, 0.3609, 0.3152 and 0.5366 rad give a pooled RMSE of 0.3583 rad (the "Overall" row) and a mean of the per-joint RMSEs of 0.3418 rad.

Numerical check: `test_mean_of_joint_rmse_never_exceeds_pooled_rmse, test_pooled_rmse_is_not_mean_of_rmse`.

The *lag* of joint j is $s^\star\Delta t$ with $s^\star = \arg\max_{0 \le s \le s_{\max}} \varrho(q_m[0{:}K{-}s],\, q_s[s{:}K])$, ϱ the Pearson correlation, at the resolution of one sample (20 ms); the *lag-compensated RMSE* is the RMSE of $q_s[s^\star{:}] - q_m[{:}K{-}s^\star]$. The table entries "free" use the first 12 s of the task and "all" the whole task.

11.2 Force and estimation

- **Force correlation** (benchmark): Pearson correlation of $f_{felt,x}$ and the true wall force over the contact samples.
- **Rendered gain** (force reproduction): the least-squares slope of the felt force on the ideal one, $g = \sum_k f_{felt}[k]\, f_{ref}[k] \,/\, \sum_k f_{ref}[k]^2$ with $f_{ref} = k_{of}F_{env}$, over the samples with $\|F_{env}\| > 0.2$ N, per axis; $g = 1$ is exact scaled reproduction. The correlation is computed over the same samples.
- **Weight rendering**: the mean vertical felt force over the last 2 s of holding a mass m, regressed on mg (slope and offset).

- **Estimation error**: RMS of $\hat{\tau}_{ext}[k+1] - \tau_{ext}[k]$ (the estimate at a tick uses the current measured at the end of the previous tick) over the loaded samples, excluding 0.3 s after each load change; the *false torque* is the RMS estimate over the unloaded samples.

Each multi-seed entry is the sample mean and standard deviation ($n-1$ normalisation) over five seeds; the seeds pair the controllers (same noise and operator for all controllers in a seed).

12 Parameter tuning in the Control Simulator

12.1 Plant and control law

The Control Simulator evaluates a free-text control law on the mass–spring–damper $m\ddot{y}+b\dot{y}+ky = u$ with $m = 1\,\text{kg}$, $b = 0.5\,\text{N s/m}$, $k = 2\,\text{N/m}$, integrated by explicit Euler with $h = 10\,\text{ms}$, $x[i+1] = x[i] + h(Ax[i] + Bu[i])$, and $|u| \le 10$. The law may read $e[i] = r[i] - y[i]$, $\mathcal{I}[i] = \sum_{j \le i} e[j]h$ and $\mathcal{D}[i] = (e[i] - e[i-1])/h$ with $e[-1] = 0$ (so that $\mathcal{D}[0] = e[0]/h$, which saturates u at the first sample of a step). For the PID law $u = K_p e + K_i\mathcal{I} + K_d\mathcal{D}$ and $h \to 0$, the closed loop without saturation is $m\ddot{y} + (b + K_d)\dot{y} + (k + K_p)y + K_i\int y = K_p r + K_i\int r + K_d\dot{r}$, whose characteristic polynomial $ms^3 + (b + K_d)s^2 + (k + K_p)s + K_i$ is Hurwitz iff all coefficients are positive and $(b + K_d)(k + K_p) > mK_i$ (Routh–Hurwitz), e.g. $0.9 \cdot 10 = 9 > 0.5$ for $K_p = 8$, $K_i = 0.5$, $K_d = 0.4$.

12.2 Objectives

For a run with samples $t_i = ih$, $i = 0, \ldots, N_s$:

$$\text{ISE} = \sum_i e_i^2 h, \qquad \text{IAE} = \sum_i |e_i| h, \qquad \text{ITAE} = \sum_i t_i |e_i| h, \qquad \text{overshoot} = 100\,\frac{\max_i y_i - r_\infty}{|r_\infty|}\,\%,$$

the rise time is the time between the first samples with $y \ge 0.1 r_\infty$ and $y \ge 0.9 r_\infty$, and the settling time is the first t_i after which $|y - r_\infty| \le 0.02|r_\infty|$ for all later samples.

12.3 Search

The parameters declared with `@param` define a box $\Theta = \prod_p [\theta_p^{\min}, \theta_p^{\max}]$. The tuner solves $\min_{\theta \in \Theta} J(\theta)$, with J one of the objectives above, by differential evolution [11] (followed by a bounded quasi-Newton polish) or by Nelder–Mead/Powell. Because J is evaluated by simulation, only the box constraints of the declaration are enforced; saturation of u makes J non-smooth, which is why a population method is the default.

13 What is not proven, and what hardware must confirm

1. **Closed-loop stability** of the MPC (no terminal ingredients) and of the full teleoperation loop, in particular with delay.
2. **Model validity on the real arm**: decoupled second-order position loop with a constant disturbance (Assumption 3); the reported ω_i, ζ_i were identified on the simulated slave. Backlash, the servo velocity/acceleration profiles and current limiting are not modelled.
3. **Torque constants and inertial parameters** (Remark 4): the observer's accuracy is limited by k_t, the link masses and friction, which must be calibrated.
4. **Energy accounting** is sampled-data (Section 8); P_{in} assumes the master's and the admittance's damping terms are implemented as written.
5. The **dead zones, gains and the horizon** were chosen in simulation.

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